

An Accurate Three-Axis Force Sensor Assembled with Readily Available Off-the-Shelf Components with No Designed Compliant Element

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Abstract—Affordable multi-axis force sensors are usually designed with a custom compliant part, and the deformation of that part is measured after a load is applied. Any creep, hysteresis, or thermal sensitivity in that part then becomes a property of the finished sensor. Its accuracy then depends on operating temperature and load history in ways a single calibration cannot capture. We instead developed a sensor using four commercial load cells that are manufacturer-characterized to carry the full load. On top of the mechanical design, one fitted matrix that maps readings to force accounts for assembly tolerances, gain differences, and wiring polarity. The resulting three-axis sensor costs under \$120 and has a 2.6–3.5 mN noise floor. Off-axis leakage remains below 1 %, and lateral hysteresis remains below 0.5 % of span. After one scale factor, its output agrees with a commercial reference to 1.0–2.5 % of full scale across more than 300,000 in-contact samples. The design suits tools whose contact point is known and makes an accurate three-axis sensor accessible to any research lab with a 3D printer.

I. INTRODUCTION

Force sensing is critical in any task where a robot meets a surface it cannot fully model. Commercial multi-axis sensors do this well but cost tens of thousands of dollars [1], which has pushed a research effort toward cheaper designs.

Those low-cost designs have used optical readout [2], cameras and fiducials [1], capacitance [3], and light transmitted through an elastomer [4]. In each case an inexpensive sensor measures a compliant structure built for that device. The structure determines range and sensitivity, and it also adds creep, hysteresis, and temperature dependence. These effects are seldom characterized in full. Hysteresis is sometimes reported, but creep and temperature dependence usually are not, so a user cannot predict the error under a sustained load, under load cycling, or at a different operating temperature.

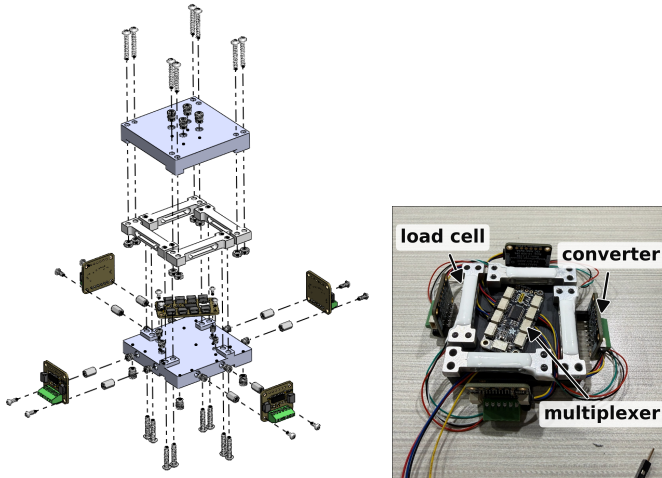


Fig. 1. The sensor as designed and as built. On the left, an exploded view of the body, with the interchangeable tip and the microcontroller omitted. On the right, the four load cells that form the structure, with the converters on the perimeter and the I²C multiplexer at the center.

The four-cell arrangement already exists in a design by Chua and Okamura, which places four commercial point-contact cells around a PCB perimeter [5]. That design needs a custom PCB, which makes it far less readily available and much more expensive. Ours keeps the cost low and the error terms quantifiable while using only off-the-shelf parts. The load cells serve as both beams and transducers. Their beam geometry, strain-gauge placement and bridge compensation are set at manufacture, and the creep and temperature behavior that follows is specified on the datasheet [6], so no separate part needs its own characterization.

II. METHOD

A. Sensor Structure

Four TAL221 500 g bar load cells connect the two 3D-printed plates in a square (Fig. 1). Each senses compression along the vertical axis. Together they respond to F_z , M_x , and M_y . The remaining combination is the self-stress mode, a pattern of internal cell forces that cannot be produced by an external load.

Each load cell has its own NAU7802 converter, interleaved through an I²C multiplexer for an aggregate output of about 1200 Hz. The multiplexer connects to a Teensy 4.0, though any reasonably capable microcontroller would serve.

A lateral force is recovered from its moment about the cell plane. For a contact at (x_c, y_c) and height h , $M_y = -F_x h + F_z x_c$ and $M_x = F_y h - F_z y_c$. The contact location must therefore be known. For the tests reported here, the contact face lies 38.6 mm above the cell mid-plane.

We tare each channel before loading to remove the assembly preload and the weight of the parts above the cells. A fitted 3×4 matrix with a bias converts a four-entry vector of tared counts into (F_x, F_y, F_z) . The same fit absorbs placement error, cell-to-cell gain variation, polarity, and cross-coupling. Firmware identifies the multiplexer channel used by each converter, so the physical cell order does not matter.

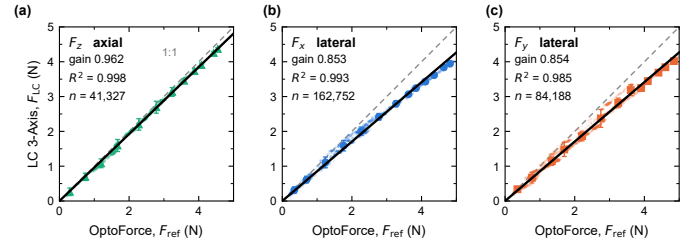


Fig. 2. Response against the OptoForce reference with both instruments in series on the same rig. Dashed gray is the 1:1 line and black is a least-squares fit through the origin. Markers are slice means with their standard deviation. The gains and R^2 shown are for the magnitude fit, which is frame-invariant. The per-axis component values quoted in the text are gains of 0.953, 0.865 and 0.873 at $r^2 \geq 0.993$. The spread below 1 N in (a) is the reference's own 0.0319 N quantization rather than sensor noise.

TABLE I
BENCH BEHAVIOR, CALIBRATED RANGE 0–4.9 N

	F_x	F_y	F_z
Leave-one-out error (N)	0.0431	0.0330	0.0357
Noise floor, 1σ (mN)	3.2	3.5	2.6
On-axis decoupling (%)	100.1	99.7	99.6
Hysteresis (% of span)	0.111	0.421	1.112

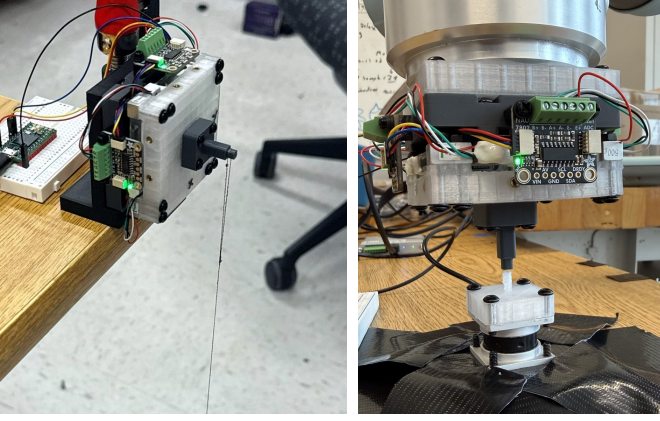


Fig. 3. The two calibrations. Left, the sensor clamped axis-horizontal at the bench edge with known masses hanging from the tip, so gravity loads it perpendicular to the sensing axis. Right, the two instruments in series on the UR5e, the sensor on the flange and the OptoForce below it, so each carries the same force.

B. Fabrication

Only three of the seventeen part types are custom. They are two FDM 3D-printed PLA plates and an interchangeable printed tip. The load cells, converters, multiplexer, and micro-controller cost \$116.55 in total. Fasteners, cabling, and filament are excluded. The bill of materials, firmware and calibration procedure are at <https://github.com/lucasfeng7/lc-3-axis>.

III. EXPERIMENT

A. Sensor Characterization

We calibrated against known masses rather than another force sensor (Fig. 3). For each lateral axis the sensor was clamped horizontal, so gravity acted perpendicular to the axis being loaded.

Table I reports leave-one-out error, which stays below 0.9 % of range on every axis. Residuals measured in sample are 2.2 times smaller. On-axis decoupling is close to a full response on every axis, and the F_x entry sits just above it because of a small overshoot in the estimate. Leakage onto the other two axes stays below 1 %. Hysteresis is 0.111 % and 0.421 % of span laterally.

We applied three overloads of 19.9–28.7 N against the 19.6 N limit. After each, the unloaded output returned to within 0.05 N of zero. A different load-cell rating changes the range without changing the design.

B. Independent Validation

For validation, the UR5e carried two independently calibrated instruments in a serial stack (Fig. 3). Their coordinate frames were aligned with an orthogonal transform. The data contain 344,424 in-contact samples. The responses and fits appear in Fig. 2.

Allowing one scale factor per axis, the readings then sit within 0.1137 N of the reference on F_x and 0.1211 N on F_y ,

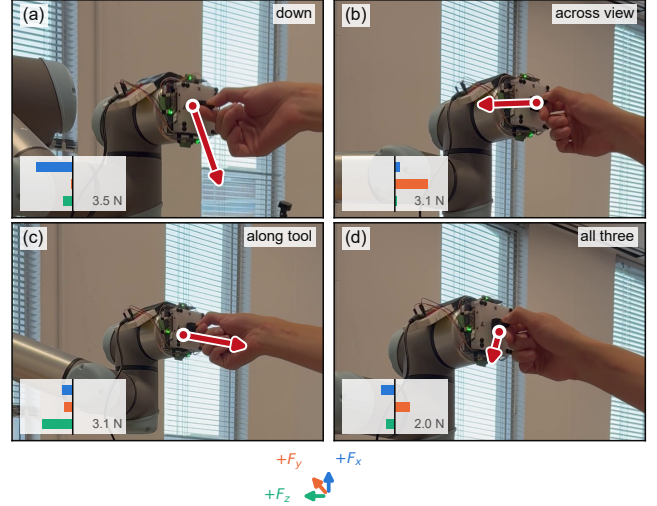


Fig. 4. The sensor driving admittance hand-guiding on the UR5e. Four instants from one run, with the components measured at each instant inset on a common 4.9 N scale, colored as the triad at the right. The red arrow is the measured force projected into the image from the contact point, so its length is foreshortened. F_y and F_z project within 1° of each other here, so the arrow gives the direction of a push but not its split between those two axes, which the bars carry. Panel (a) is a push down, (b) across the view, (c) along the tool, and (d) loads all three axes at once.

which is 2.32 % and 2.47 % of the 4.9 N range. The two rise together in a straight line, at $r^2 \geq 0.993$. For F_z we pressed in steps of increasing force along the tool axis, where the readings sit within 0.0480 N, or 0.98 %, at $r^2 = 0.998$. The scale factors themselves are 0.865 and 0.873 laterally and 0.953 axially. What separates them is scale, not scatter and not nonlinearity. The likely cause is that the robotic arm was controlled manually during the comparison, which varies the contact height and therefore the lever arm.

Even so, these figures match or exceed the closest comparable design. Both are the RMS disagreement with an independent reference after per-axis calibration constants are fitted. Ours are 0.1137 and 0.1211 N laterally and 0.0480 N axially. Chua and Okamura report 0.105 to 0.146 N laterally and 0.064 to 0.126 N axially, the spread being across the two units they built [5].

C. Sensor Applications

Finally, we used the sensor for admittance hand-guiding. In this control mode, the measured contact force commands the robot's Cartesian velocity. The loop runs at 199.8 Hz (Fig. 4). It responds to single-axis pushes and to simultaneous loading of all three axes, which confirms that the sensor is fast and accurate enough to close a control loop on a moving robot rather than only on a bench.

IV. CONCLUSION

In this study we presented a three-axis force sensor which leverages four load cells as the structure. The sensor does not need a custom PCB and is simple to assemble, and it agrees with a commercial reference to 1.0–2.5 % of full scale. It matches or exceeds the accuracy of the closest comparable design, which makes it an affordable and easy-to-implement option for research labs that need accurate three-axis force sensing.

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